Exam Date & Time: 27-Jul-2022 (09:00 AM - 12:00 PM)

MANIPAL ACADEMY OF HIGHER EDUCATION

FIRST SEMESTER B.TECH MAKE-UP EXAMINATIONS, JULY 2022

DIGITAL IMAGE PROCESSING [BME 3252]

Marks: 50 **Duration: 180 mins.**

A

Answer all the questions.

Instructions to Candidates:
Answer ALL questions Missing data may be suitably assumed

1) Consider the system with input-output relationship: $y(m,n) = x^2(m, n)$. Find out if the system is linear, shift-invariant, and causal. Justify each of your answers mathematically. (4)

A)

B) It is often sufficient to know the working of a 1D system, to understand that of its counterpart on an image. Accordingly, consider the system with input-output relationship:
$y(n) = (1/3)\,[\,-x(n-1) + 2x(n) - x(n+1)]$.
Find the expression for the frequency-response of the filter, and plot its magnitude. (3)

C) Consider filtering a 2D sequence of size 610×760. The size of the filtering kernel is 50×50. Find the number of zeros to be padded to the two sequences, to allow linear filtering by the kernel, using an FFT algorithm (3)

2) Explain the experiment to assess the *just-noticeable difference* associated with a subject. What is the conclusion of the experiment? What would be the results, if you allow infinitesimal (continual) increments in intensity – explain mathematically. (5)

A)

B) Compute the 3 X 3 Discrete Cosine Transform (DCT) matrix, withe numerical entries, given the following expression for the DCT:

$$C_x(k) = v(k)\sum_{n=0}^{N-1} x(n)\cos\left[\frac{\pi}{2N}(2n+1)k\right], \qquad 0 \le k \le N-1$$

where: $v(0) = \sqrt{\frac{1}{N}}, \qquad v(k) = \sqrt{\frac{2}{N}}, \qquad 1 \le k \le N-1$

(5)

Compute the inverse of the DCT matrix with numerical values.

3) Compute the output of the 5-point averaging filter i.e., the filter whose mask is given by: (4)

A)

0.0	0.2	0.0
0.2	0.2	0.2

0.0	0.2	0.0

on the following image (Fig. 1). Show the calculations any three important locations.

8	8	8	8	20	20	10	20	20
8	8	8	8	20	20	10	20	20
8	8	28	8	20	20	10	20	20
8	8	8	8	20	20	10	20	20
20	20	20	20	20	20	10	20	20
20	20	20	20	20	20	10	20	20
20	20	20	20	20	20	10	20	20
20	20	20	20	20	20	10	20	20

Figure 1.

B) Find the output of a 5-point median filter i.e., a median filter with a cross-shaped support as shown in the following:

	x	
x	x	x
	x	

(4)

on the image in Fig. 1 (previous question). Show the calculations at any 3 important locations.

C) Compare the results averaging and median filtering, by pointing out the important regions in the two output images. (2)

4) Develop from fundamentals, the gradient-based edge-detection scheme. (5)

A)

B) Write the complete *pseudo-code* to detect the presence of parabolae of the form: $ax^2 + bx + c = 0$ in digital images. (5)

5) Find the result of closing a circular disc by itself. You must sketch all the intermediate results, clearly. (4)

A)

B) Explain the application of morphological operations for finger-print enhancement. (4)

C) What are the desirable properties of feature-vectors for successful classification? (2)

-----End-----